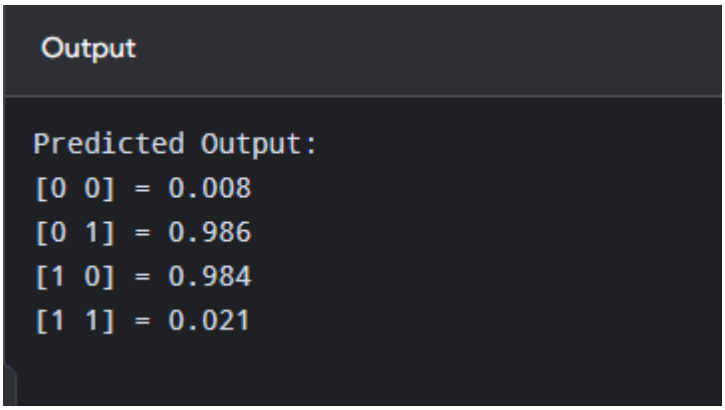
Lab experiment

Feed Forward Neural Network

Coding:

```
# Feed Forward Neural Network using Python

import numpy as np

# Input data (XOR problem)
X = np.array([
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
])

# Expected output
y = np.array([
    [0],
    [1],
    [1],
    [0]
])

# Initialize weights and biases
np.random.seed(1)

W1 = np.random.randn(2, 4)
b1 = np.zeros((1, 4))

W2 = np.random.randn(4, 1)
b2 = np.zeros((1, 1))

# Sigmoid activation function
def sigmoid(x):
    return 1 / (1 + np.exp(-x))

# Training
learning_rate = 0.5

for epoch in range(10000):
```

```
    # Forward propagation
    hidden = sigmoid(np.dot(X, W1) + b1)
    output = sigmoid(np.dot(hidden, W2) + b2)

    # Calculate error
    error = y - output

    # Backpropagation
    delta_output = error * output * (1 - output)

    delta_hidden = (
        np.dot(delta_output, W2.T)
        * hidden * (1 - hidden)
    )

    # Update weights and biases
    W2 += learning_rate * np.dot(
        hidden.T, delta_output
    )

    b2 += learning_rate * np.sum(
        delta_output, axis=0, keepdims=True
    )

    W1 += learning_rate * np.dot(
        X.T, delta_hidden
    )

    b1 += learning_rate * np.sum(
        delta_hidden, axis=0, keepdims=True
    )

# Display final output
print("Predicted Output:")

for i in range(len(X)):
    hidden = sigmoid(np.dot(X[i:i+1], W1) + b1)
    output = sigmoid(np.dot(hidden, W2) + b2)

    print(X[i], "=", round(output[0][0], 3))
```

Output: